

/return executed within an advancement reward function globally discards all subsequent commands

MC-266968 Advancements

Status	Resolved / Fixed
Affected	1.20.3 Pre-Release 2
Fixed in	1.20.3 Pre-Release 4
Reported	2023-11-24

STEPS TO REPRODUCE

1. Open a Minecraft world.
2. Install the data pack from the report's Attachments into the world.
3. Run `/function mc-266968:a``.
4. Observe the chat output: ``a-1`` and ``b-1`` are printed, while ``a-2`` and ``b-2`` are not printed.

OBSERVED BEHAVIOUR

Actual behavior

Running `/function mc-266968:a`` prints:

```
```text
```

```
a-1
```

```
b-1
```

```
```
```

The ``return 0`` command in ``mc-266968:b``, which is executed as the reward function for the granted advancement, prevents both of the following commands from executing:

- ``say b-2`` in ``mc-266968:b`` - expected, because it follows ``return 0`` in the same function.
- ``say a-2`` in ``mc-266968:a`` - unexpected, because it follows the command that granted the advancement and belongs to the caller function.

Thus, ``return 0`` in an advancement reward function does not only terminate that reward function. It also discards the pending commands in the calling function, causing subsequent commands to be skipped globally.

EXPECTED BEHAVIOUR

When the advancement reward function executes ``return 0``, it should stop only the remaining commands within that reward function. The surrounding function should continue normally, so ``b-1`` should be printed, ``b-2`` should not be printed, and execution should then return to ``mc-266968:a``, where ``a-2`` should also be printed. Subsequent commands should not be discarded globally.

ENVIRONMENT

Minecraft Java Edition Version 1.20.3 Pre-Release 2

ORIGINAL DESCRIPTION

The bug

When `/return` is executed within an advancement reward function, it will discard all subsequent commands globally, not just the local commands that the function has.

How to reproduce

- Install the data pack in Attachments
- Run

```
/function mc-266968:a
```

Results

The following table shows whether each message was expected to be printed or not, and whether it was actually printed or not.

Message

Expected

Actual

a-1

Yes

Yes

a-2

Yes

No

b-1

Yes

Yes

b-2

No

No

As the table suggests, say a-2 in mc-266968:a was not executed because it was discarded by return 0 in mc-266968:b.

Resources

data/mc-266968/functions/a.mcfuction

say a-1

advancement grant @s only mc-266968:b

say a-2

data/mc-266968/functions/b.mcfuction

advancement revoke @s only mc-266968:b

say b-1

return 0

say b-2

data/mc-266968/advancements/b.json

```
{
  "criteria": {
    "": {
      "trigger": "impossible"
    }
  },
  "rewards": {
    "function": "mc-266968:b"
  }
}
```

Code analysis

An advancement reward function is executed as an initial function, which has a frame with depth 0, regardless of the current depth. Because /return discards all command queue entries above the depth of the currently executed function, /return executed in an advancement reward function discards all entries, which have a depth ≥ 0 .

MANUAL RESULT

| | |
|------------------|--------------------|
| Version used | |
| Reproduced? | YES / NO / PARTIAL |
| Crash file | |
| Notes | |
| | |
| Tested by / date | |

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-266968>